Circuit Analysis (Lab3)

Experiment 3: Introduction to Multimeter and Basic Series Circuit Implementation

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Date: 18-March-2026 **Grade and Signature: ………………………..**

CLO2 **CLO3**	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	**Level1 (1)**	**Level2 (2-3)**	**Level3 (4-5)**	**Level4 (6-7)**
Report Marks (3)			**Total marks (10)**	

Objectives: To get understanding of how to use a multimeter and make measurements in your own constructed series circuits.
Equipment required:
Power Supply, Breadboard, Connecting wires, Resistances, Multimeter.

What is a multimeter?

A multimeter is a measurement tool necessary in electronics. It combines three essential features: a voltmeter, ohmmeter, and ammeter, and in some cases continuity. A multimeter allows you to understand what is going on in your circuits. Whenever something in your circuit isn't working, the multimeter will help you troubleshooting.

Parts of Multimeter:

Display: this is where the measurements are displayed
Selection knob: this selects what you want to measure
Ports: this is where you plug in the probes
Probes: a multimeter comes with two probes. Generally, one is red and the other is black.
Note: There isn't any difference between the red and the black probes, just the color. So, assuming the convention:

- The black probe is always connected to the COM port.
- The red probe is connected to one of the other ports depending on what you want to measure.

Ports

The "**COM**" or "–" port is where the black probe should be connected. The COM probe is conventionally black.

- **A** is used when measuring large currents, greater than 200mA
- **mA** is used to measure current
- **VΩ** allows you to measure voltage and resistance and test continuity.

Figure 1 Digital Multimeter (DMM)

Introduction to resistance measurements

Resistance is measured with the circuit's power turned off. The ohmmeter sends its own current through the unknown resistance and then measures that current to provide a resistance value readout.

Resistance measurement procedures

Follow the steps below to measure resistance:

1. Turn off the power to the circuit. Remove or isolate the component to be tested.
2. Plug the test probes into the appropriate probe jacks. Note that the jacks used may be the same ones used to measure volts.
3. Select the ohms function by turning the function switch to ohms. Start with the lowest setting.
4. Touch the probes together to check the leads, connections and battery life. The meter should display zero or a very small amount of resistance for the test leads. With the leads apart, the meter should display OL or I, depending on the manufacturer.
5. Connect the tips of the probes across the break in the component or portion of the circuit for which you want to determine resistance. If you get an OL (over limit), move to the next level.
6. View the reading on the display unit. Be sure to note the unit of measurement.
7. After all the resistance readings are complete, turn the DMM off to prevent the battery from draining

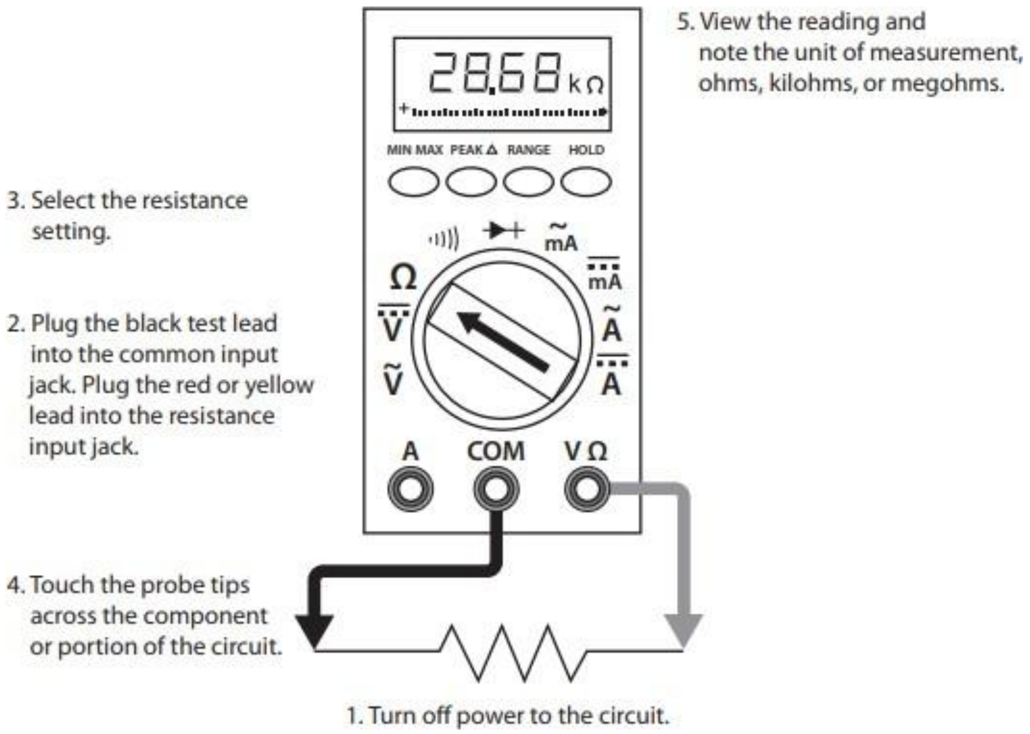


Figure 2 Using a DMM to measure resistance

Introduction to voltage measurements

Voltage measurements are very easy to make with a DMM. On meters with manual range selection, start with a value one setting higher than expected. An auto range DMM will automatically select the range based on the voltage present. Figure 4 shows the process.

Note:

1/ 1000 V=1mV

1000 V=1kV

Follow these steps to measure voltage:

1. Select the DC or AC volts function by turning the function switch to DC or AC volts.
2. Plug the test probes into the appropriate probe jacks.
3. Connect the tips of the probes across the source or load.
4. View the reading on the display unit. Be sure to note the unit of measurement. If you are testing DC voltage and a negative sign appears in the display, the polarity of your probes is incorrect and needs to be reversed.

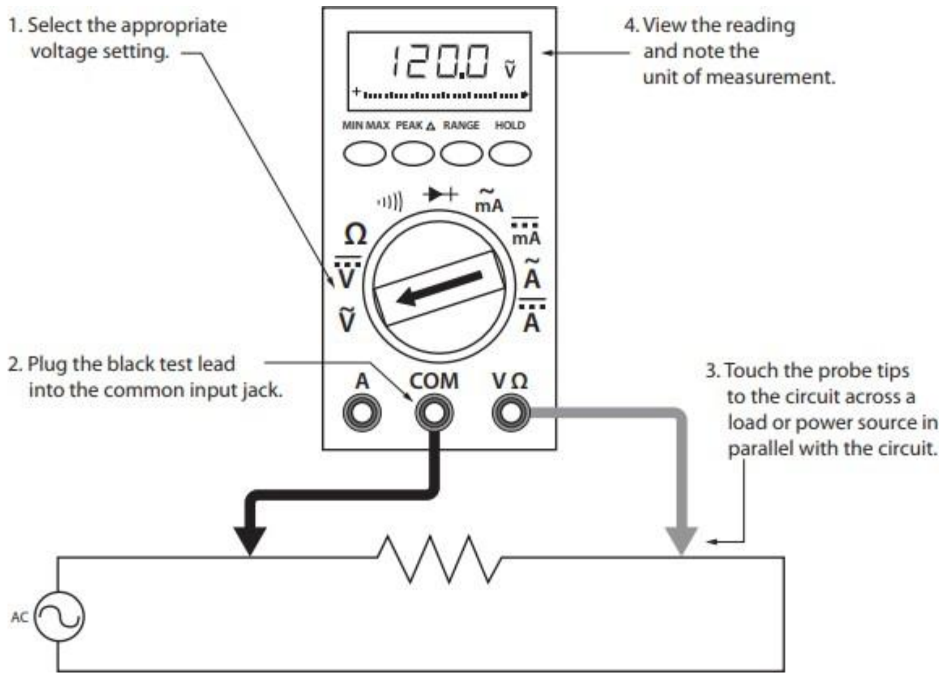


Figure 3 Using a DMM to measure voltage

Introduction to current measurements

As you have seen, the placement of meter leads for voltage measurements is straightforward. The leads are simply connected across, or in parallel with, the points of voltage to be measured. For current measurements, however, the process is slightly more complex.

First, the circuit must be opened at the test points and the meter inserted in series at that opening. The total current must flow through the meter. To allow the measurement to be made without disturbing the circuit itself, the current meter must have very little internal resistance.

If the meter is inadvertently connected across a P.D. (potential difference) or in parallel with a component instead of in series, the small internal resistance will permit a very large current to flow through the meter. This will most certainly damage the meter severely and perhaps the circuit as well.

On meters with manual range selection, start with the highest current setting and work your way down.

Current measurement procedures

Follow the steps below to measure current:

1. Turn off the power to the circuit that will be measured.
2. Open the circuit by disconnecting or unsoldering a connection at a point where you wish to measure current.
3. Select the DC or AC amps function by turning the function switch to DC or AC amps.
4. Plug the test probes into the appropriate probe jacks. Note that the jacks used may not be the same ones used to measure volts.
5. Connect the tips of the probes across the break in the circuit so that the current to be measured flows through the meter. Note that this is a series connection. Never connect the ammeter in parallel with the source or load, as this will cause a short circuit and damage the meter.
6. Turn the circuit power back on.
7. View the reading on the display unit. Be sure to note the unit of measurement.
8. Switch power off and disconnect meter leads from the circuit.

9. If testing is finished at this test point, restore the circuit by reclosing the connection. When current measurements are complete, turn the function switch to the OFF position and remove the test leads.

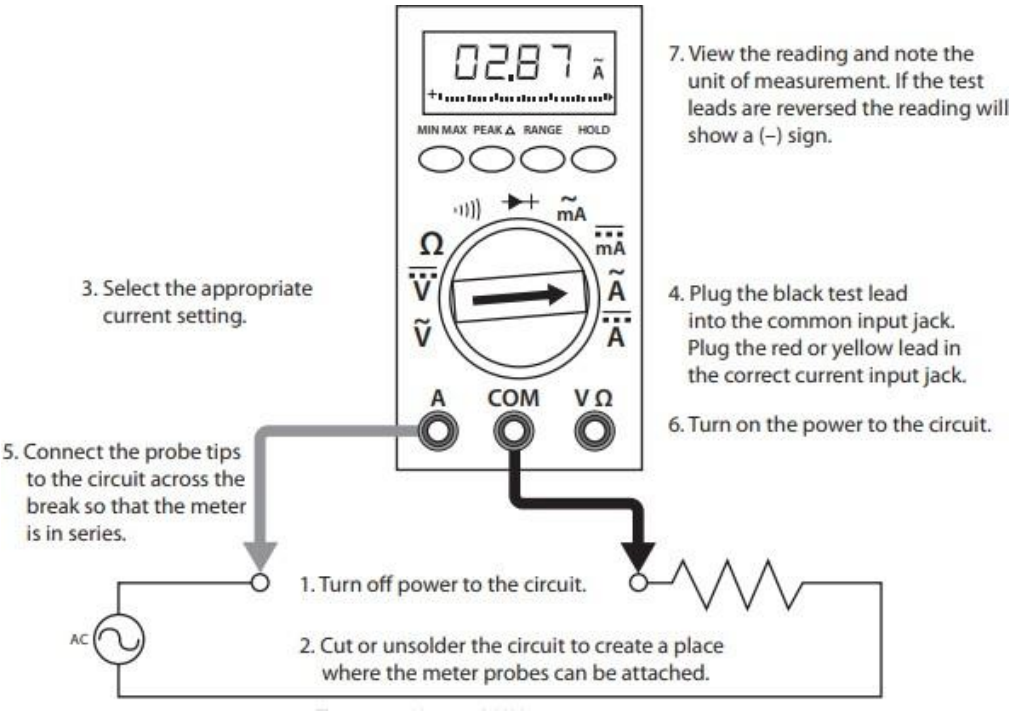
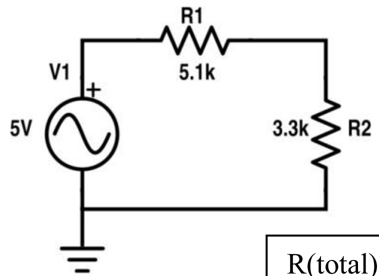


Figure 4 Using a DMM to measure current

Lab Tasks:

1. Measure Current between R1 and R2, and Voltage across R1, R2.



R(total) = 8.4k ohm I = 5/8400 = 0.595 mA

V(R1) = I x R1 = 0.595mA x 5.1k = 3.03 V
V(R2) = I x R2 = 0.595 mA x 3.3k = 1.96 V

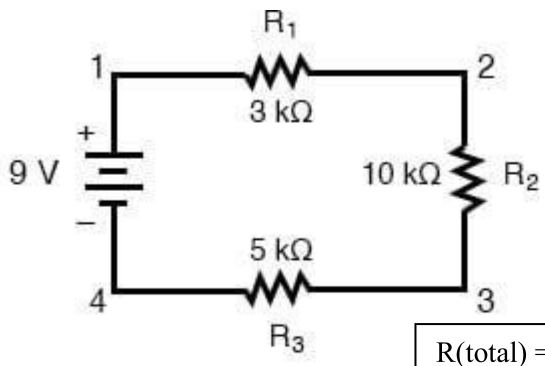
Practical Values:

R1 Voltage	R2 Voltage	Current
3.03 V	1.96 V	0.59 mA

Theoretical Values:

R1 Voltage	R2 Voltage	Current
3.03 V	1.96 V	0.595 mA

2. **Measure Current at node 2 and 3 and Voltage across R1, R2, R3.**



R(total) = 18k ohm
I = 9/18000 = 0.5 mA
V(R1) = 0.5 mA x 3k = 1.5V
V(R2) = 0.5 mA × 10k = 5V
V(R3) = 0.5 mA x 5k = 2.5V

Practical Values:

R1 Voltage	**R2 Voltage**	**R3 Voltage**	**Current at node 2**	**Current at node 3**
1.5 V	5 V	2.5 V	0.5 mA	0.5 mA

Theoretical Values:

R1 Voltage	**R2 Voltage**	**R3 Voltage**	**Current at node 2**	**Current at node 3**
1.5 V	5V	2.5 V	0.5 mA	0.5 mA

Post-Lab Questions

1. Define energy

Energy is simply the ability or capacity to do work. In our circuits, it’s the "fuel" that allows electricity to move and perform tasks, like lighting up a bulb or running a motor. It can’t be created or destroyed, it just changes from one form to another.

2. Define power

Power is the rate at which energy is being used or work is being done. While energy is the total amount of "stuff" used, power tells us how fast we are using it. In electricity, we calculate it by multiplying the voltage by the current.

$$P = V \times I$$

3. What is charge?

Charge is a basic property of matter, specifically of the tiny particles like electrons (which have a negative charge) and protons (which have a positive charge). When these charges move through a wire, we get what we call an electric current.

4. What is circuit?

A circuit is a complete, closed path or loop that electricity flows through. It usually includes a power source (like a battery), wires, and something to use the electricity (like a resistor). If the loop is broken, the electricity stops flowing.

5. What is the role of a resistor in the series circuit?

The main job of a resistor is to limit or "resist" the flow of current so it doesn't get too high. In a series circuit, resistors also create "voltage drops," which means they share the total voltage coming from the battery based on their size.

6. What is the voltage divider rule?

This is a handy math shortcut used in series circuits. It lets you figure out the voltage across one specific resistor without having to measure it. The rule says that the voltage is split up based on the ratio of that resistor to the total resistance:

$$V_R = V_{total} \times (R / R_{total})$$

7. What are the features of the series circuit?

Series circuits have a few main rules:

- The **current** is exactly the same at every single point in the loop.

- The **total resistance** is just all the individual resistors added together.
- The **total voltage** of the battery is shared or "divided" between all the components.

8. Differentiate between the internal circuit of voltmeter and ammeter

The main difference is their internal resistance:

- An **ammeter** has very **low resistance**. This is so it doesn't block the current it's trying to measure.
- A **voltmeter** has very **high resistance**. This is so it doesn't "steal" any current from the circuit while it's checking the voltage.

9. In which combination we connect DMM to measure current and voltages?

- **For Voltage:** You connect the DMM in **parallel** (placing the probes across the component without breaking the wire).
- **For Current:** You connect the DMM in **series** (you have to physically break the circuit and make the electricity flow through the meter).